

Tutorial 3 - Q2

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Question 2 By considering the existence of moments show that the process

$$Y_t = Z_t + a(Z_{t-1} + Z_{t-2} + \dots)$$

where a is a constant, is non-stationary. Show, however, that the process $\{V_t\}$ obtained by taking first differences, i.e. $V_t = DY_t = Y_t - Y_{t-1}$, is an $MA(1)$ process and hence stationary. Calculate the autocorrelation function of $\{V_t\}$.

We have the following process :

$$\begin{aligned} Y_t &= Z_t + a(Z_{t-1} + Z_{t-2} + \dots) \\ &= Z_t + aZ_{t-1} + aZ_{t-2} + \dots \end{aligned}$$

Stationary conditions:

$$\begin{aligned} \textcircled{1} \quad E[Y_t] &= E[Z_t + aZ_{t-1} + \dots] \\ &= E[Z_t] + aE[Z_{t-1}] + \dots \\ &= 0, \text{ since } E[Z_t] = 0 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad \text{Var}[Y_t] &= \text{Var}[Z_t + aZ_{t-1} + aZ_{t-2} + \dots] \\ &= \text{Var}[Z_t] + a^2 \text{Var}[Z_{t-1}] + \dots \\ &= \sigma_Z^2 + a^2 \sigma_Z^2 + \dots \\ &= \sigma_Z^2 (1 + a^2 + \dots) \rightarrow \infty \text{ as } t \rightarrow \infty \end{aligned}$$

So condition $\textcircled{2}$ fails $\Rightarrow$ non-stationary.

Now let's define V_t

$$\begin{aligned} V_t &= DY_t \\ &= Y_t - Y_{t-1} \\ &= [Z_t + a(Z_{t-1} + Z_{t-2} + \dots)] - [Z_{t-1} + a(Z_{t-2} + Z_{t-3} + \dots)] \\ &= Z_t + aZ_{t-1} - Z_{t-1} \end{aligned}$$

$$V_t = Z_t + (a-1)Z_{t-1}$$

Stationarity conditions:

$$(i) \quad E[V_t] = \underbrace{E[Z_t]}_0 + (a-1) \underbrace{E[Z_{t-1}]}_0$$

$\therefore$ mean is constant for all t .

(ii)

$$\begin{aligned} \text{Var}[V_t] &= \text{Var}[Z_t + (a-1)Z_{t-1}] \\ &= \text{Var}[Z_t] + (a-1)^2 \text{Var}[Z_{t-1}] \\ &= \sigma_z^2 + (a-1)^2 \sigma_z^2 \end{aligned}$$

$\therefore$ variance is constant for all t .

$$\begin{aligned} (iii) \quad \gamma(\pm 1) &= \text{Cov}(Z_t + (a-1)Z_{t-1}, Z_{t-1} + (a-1)Z_{t-2}) \\ &= \text{Cov}(Z_t, Z_{t-1}) + \text{Cov}(Z_t, (a-1)Z_{t-2}) \\ &\quad + \text{Cov}((a-1)Z_{t-1}, Z_{t-1}) + \text{Cov}((a-1)Z_{t-1}, (a-1)Z_{t-2}) \\ &= (a-1)\text{Var}(Z_{t-1}) \\ &= (a-1)\sigma_z^2 \end{aligned}$$

$\therefore V_t$ is stationary as $\gamma(\pm 1)$ is indep of t .

We have the following ACF,

$$\rho(\pm k) = \begin{cases} 1 & , \text{ for } k=0 \\ \frac{a-1}{1+(a-1)^2} & , \text{ for } |k|=1 \\ 0 & , \text{ for } |k| \geq 2 \end{cases}$$